

1. Solve:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

$$u|_{x=0} = u_0, \quad u|_{x=a} = u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right],$$

$$\frac{\partial u}{\partial y} \Big|_{y=0} = 0,$$

$$\frac{\partial u}{\partial y} \Big|_{y=b} = 0,$$

(b=constant)

$$u(x,y) = X(x)Y(y) \Rightarrow X''(x)Y(y) + X(x)Y''(y) = 0, \quad \begin{cases} X''(x) - \lambda X(x) = 0 \\ Y''(y) + \lambda Y(y) = 0 \end{cases} \quad \frac{du}{dy} \Big|_{y=0} = 0, \quad \frac{du}{dy} \Big|_{y=b} = 0$$

$$\text{当 } \lambda = 0 \quad Y''(y) = 0 \Rightarrow Y(y) = Ay + B \quad \text{代入边界 } A = 0 \text{ 不满足}$$

$$\text{当 } \lambda < 0 \quad \lambda = -s^2 \quad s = \sqrt{-\lambda}. \quad Y(y) = Ae^{sy} + Be^{-sy} \quad Y'(y) = As e^{sy} - Bs e^{-sy} \quad Y'(0) = As - Bs = 0 \quad A = B \quad Y(b) = As(e^{sb} - e^{-sb}) = 0 \quad A:$$

$$\text{不满足} \quad \text{当 } \lambda > 0 \quad \lambda = \alpha^2 \quad Y(y) = A \cos \alpha y + B \sin \alpha y \quad \text{则有 } Y'(0) = B\alpha = Y'(b) = -A\alpha \sin \alpha b + B\alpha \cos \alpha b = 0 \Rightarrow B = 0 \quad \text{则 } A\alpha \sin \alpha b = 0$$

$$\text{为使 } Y(y) = 0 \quad \text{则 } A \neq 0, \sin \alpha b = 0. \quad \text{可得 } \alpha = \left(\frac{n\pi}{b} \right)^2. \quad \text{且 } Y_n(y) = A_n \cos \left(\frac{n\pi y}{b} \right) \quad (n=1, 2, \dots)$$

$$\text{则代入有 } X'(x) - \left(\frac{n\pi}{b} \right)^2 X(x) = 0 \quad X_n(x) = C_n \sinh \left(\frac{n\pi x}{b} \right) + D_n \cosh \left(\frac{n\pi x}{b} \right)$$

$$\text{则有一般解 } u(x,y) = \sum_{n=1}^{\infty} [C_n \sinh \left(\frac{n\pi x}{b} \right) + D_n \cosh \left(\frac{n\pi x}{b} \right)] \cos \left(\frac{n\pi y}{b} \right) \quad \text{代入 } \begin{cases} X(0)Y(y) = u_0 \\ X(a)Y(y) = u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right] \end{cases}$$

$$\Rightarrow \begin{cases} \sum_{n=1}^{\infty} C_n \cos \frac{n\pi y}{b} = D_n = u_0 \\ \sum_{n=1}^{\infty} \left(C_n \sinh \frac{n\pi a}{b} + D_n \cosh \frac{n\pi a}{b} \right) \cos \frac{n\pi y}{b} = u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right] \end{cases} \quad \text{利用傅里叶级数正交性, } D_n = \frac{2}{b} \int_0^b u_0 \cos \frac{n\pi y}{b} dy.$$

$$\begin{cases} X_n''(x) - \lambda_n X_n(x) = 0 \\ X_n(0) = a_n, \quad X_n(a) = b_n \end{cases} \Rightarrow a_n = \frac{\int_0^b u_0 \cos \frac{n\pi y}{b} dy}{\int_0^b \cos^2 \frac{n\pi y}{b} dy} = 0 \quad b_n = \frac{\int_0^b u_0 \left[3 \left(\frac{y}{b} \right)^2 - 2 \left(\frac{y}{b} \right)^3 \right] \cos \frac{n\pi y}{b} dy}{\int_0^b \cos^2 \frac{n\pi y}{b} dy} = \begin{cases} 0, n \text{ 为偶} \\ -\frac{48u_0}{\lambda_n^2 b^4}, n \text{ 为奇} \end{cases}$$

$$\text{则 } X_n = \begin{cases} -\frac{48u_0}{(\frac{n\pi}{b})^4 \sinh \frac{n\pi a}{b}} \sinh \frac{n\pi x}{b}, n \text{ 为奇} \\ 0, n \text{ 为偶} \end{cases} \Rightarrow u(x,y) = u_0 \left(1 - \frac{x}{2a} \right) - \frac{48u_0}{\pi^4} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^4} \frac{\sinh \frac{2n+1}{2} \frac{\pi x}{b}}{\sinh \frac{2n+1}{2} \frac{\pi a}{b}} \cos \frac{2n+1}{2} \frac{\pi y}{b}.$$

2. Solve:

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = bx(l-x),$$

$$u|_{x=0} = 0, \quad u|_{x=l} = 0,$$

$$u|_{t=0} = 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = 0.$$

设方程一个特解为 $v(x)$, 则 $v'' = -\frac{b}{a^2} x(l-x)$, 使之满足齐次边界条件, 解之得

$$v = \frac{b}{12a^2} x(x^3 - 2lx^2 + l^3). \quad \text{设 } u = v + w, \quad \text{则 } w \text{ 满足}$$

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, w|_{x=l} = 0 \\ w|_{t=0} = -\frac{b}{12a^2} x(x^3 - 2lx^2 + l^3), \frac{\partial w}{\partial t} \Big|_{t=0} = 0 \end{cases}.$$

$$w(x,t) = \sum_{n=1}^{\infty} \left(A_n \sin \frac{n\pi}{l} at + B_n \cos \frac{n\pi}{l} at \right) \sin \frac{n\pi}{l} x,$$

$$A_n = 0, \quad B_n = -\frac{b}{6a^2 l} \int_0^l x(x^3 - 2lx^2 + l^3) \sin \frac{n\pi}{l} x dx = \frac{4l^4 b}{n^5 \pi^5 a^2} [(-1)^n - 1],$$

$$\text{所以 } u(x,t) = \frac{b}{12a^2} x(x^3 - 2lx^2 + l^3)$$

$$- \frac{8l^4 b}{\pi^5 a^2} \sum_{n=1}^{\infty} \frac{1}{(2k+1)^5} \cos \frac{(2k+1)\pi}{l} at \sin \frac{(2k+1)\pi}{l} x.$$

3. Solve in rectangular region: $0 \leq x \leq a, -b/2 \leq y \leq b/2$

$$(1) \begin{cases} \nabla^2 u = -2, \\ u|_{x=0,a} = 0, \\ u|_{y=\pm b/2} = 0; \end{cases}$$

$$(2) \begin{cases} \nabla^2 u = -x^2 y, \\ u|_{x=0,a} = 0, \\ u|_{y=\pm b/2} = 0. \end{cases}$$

(1) 设 $u(x,y) = V(x) + W(y)$ 其中 $V(x)$ 为特解 $\frac{d^2 V}{dx^2} = -2$, $V|_{x=0,a} = 0$ 易求得

$$V(x) = -x^2 + ax$$

又有 $\frac{d^2 W}{dy^2} = 0$ 求得一般解 $W(y) = \sum_{n=1}^{\infty} (C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}}) \sin \frac{n\pi x}{a}$, 代入 $W|_{y=\pm b/2} = -V(x)$ 可得

$$W|_{y=0,a} = 0 \quad W|_{y=\pm b/2} = -V(x)$$

$$C_n e^{\frac{n\pi b}{2a}} + D_n e^{\frac{n\pi b}{2a}} = A_n \Rightarrow D_n = \frac{\sinh(\frac{n\pi b}{2a})}{\sinh(\frac{n\pi b}{a})} A_n, \quad C_n = \frac{\sinh(\frac{n\pi b}{2a})}{\sinh(\frac{n\pi b}{a})} A_n$$

$$C_n e^{\frac{n\pi b}{2a}} + D_n e^{-\frac{n\pi b}{2a}} = A_n \Rightarrow A_n = \frac{\cosh(\frac{n\pi b}{2a})}{\cosh(\frac{n\pi b}{a})} A_n$$

则可求得 $u(x,y) = -x^2 + ax - \frac{8a^2}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^3} \frac{\cosh(\frac{2n+1}{2} \frac{\pi y}{a})}{\cosh(\frac{2n+1}{2} \frac{\pi y}{a})} \sin \frac{2n+1}{2} \frac{\pi x}{a}$

(2) 同理令 $u(x,y) = W(x,y) + V(x,y)$, 可求得 $V(x,y) = -\frac{1}{12} xy(x^2 - a^2)$

对 $W(x,y)$ 有 $\frac{d^2 W}{dx^2} + \frac{d^2 W}{dy^2} = 0$

$$W|_{x=0,a} = 0 \quad W|_{y=\pm b/2} = -V(x,y)$$

$$W(x,y) = \sum_{n=1}^{\infty} (A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}}) \sin \frac{n\pi x}{a}$$

$$W(x, \frac{b}{2}) = \sum_{n=1}^{\infty} (A_n e^{\frac{n\pi b}{2a}} + B_n e^{-\frac{n\pi b}{2a}}) \sin \frac{n\pi x}{a} = -\frac{bx}{24} (x^2 - a^2)$$

$$W(x, -\frac{b}{2}) = \sum_{n=1}^{\infty} (A_n e^{-\frac{n\pi b}{2a}} + B_n e^{\frac{n\pi b}{2a}}) \sin \frac{n\pi x}{a} = -\frac{bx}{24} (x^2 - a^2)$$

将两式相加得 $\frac{bx}{24} (x^2 - a^2) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{a}$, $C_n = \frac{2}{a} \int_0^a \frac{bx}{24} (x^2 - a^2) \sin \frac{n\pi x}{a} dx = \frac{2ab}{n^3 \pi^3} (\frac{1}{2} (-1)^n) - \frac{2ab}{n^3 \pi^3} [(-1)^n - 1]$

$$\begin{cases} A_n e^{\frac{n\pi b}{2a}} + B_n e^{-\frac{n\pi b}{2a}} = C_n \\ A_n e^{-\frac{n\pi b}{2a}} + B_n e^{\frac{n\pi b}{2a}} = -C_n \end{cases} \Rightarrow \text{解得 } A_n = B_n = \frac{\cosh(\frac{n\pi b}{2a})}{\sinh(\frac{n\pi b}{a})} C_n$$

则可求得 $u(x,y) = -\frac{1}{12} xy(x^2 - a^2) + \frac{ab}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \frac{\sinh(\frac{n\pi y}{a})}{\sinh(\frac{n\pi b}{2a})} \sin \frac{n\pi x}{a} + \frac{4ab}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n+1)^3} \frac{\sinh(\frac{2n+1}{2} \frac{\pi y}{a})}{\sinh(\frac{2n+1}{2} \frac{\pi y}{a})} \sin \frac{2n+1}{2} \frac{\pi x}{a}$

另外的解法:

证明. Let $u(x,y) = X(x)Y(y)$. Then the original equation becomes

$$\begin{cases} X''(x)Y(y) + X(x)Y''(y) = -2 \\ X(0)Y(y) = X(a)Y(y) = 0 \\ X(x)Y(\frac{b}{2}) = X(x)Y(-\frac{b}{2}) = 0. \end{cases}$$

So we can consider the eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(a) = 0, \end{cases}$$

which has the solution $X_n(x) = \sin \sqrt{\lambda_n} x$ with $\lambda_n = \left(\frac{n\pi}{a}\right)^2$ ($n \in \mathbb{N}$). Suppose $u(x, y) = \sum_{n=1}^{\infty} X_n(x) Y_n(y)$. Then by the orthogonality of $\{X_n(x)\}_{n=1}^{\infty}$, we have

$$\begin{cases} Y_n''(y) - \lambda_n Y_n(y) = \frac{4}{n\pi} [(-1)^n - 1] \\ Y_n(-\frac{b}{2}) = Y_n(\frac{b}{2}) = 0. \end{cases}$$

To find the expression for $Y_n(y)$, we note the Green's function associated with $Y_n(y)$ satisfies

$$\begin{cases} \frac{d^2}{dy^2} g(y; t) - \lambda_n g(y; t) = \delta(y - t) \\ g(-\frac{b}{2}; t) = g(\frac{b}{2}; t) = 0. \end{cases}$$

Therefore

$$g(y; t) = \begin{cases} A(t) \sinh \sqrt{\lambda_n} (y + \frac{b}{2}), & -\frac{b}{2} < y < t \\ B(t) \sinh \sqrt{\lambda_n} (y - \frac{b}{2}), & t < y < \frac{b}{2}, \end{cases}$$

where the coefficient function $A(t)$ and $B(t)$ are determined by the continuity property of $g(y; t)$ at $y = t$:

$$\begin{cases} A(t) \sinh \sqrt{\lambda_n} (t + \frac{b}{2}) = B(t) \sinh \sqrt{\lambda_n} (t - \frac{b}{2}) \\ \sqrt{\lambda_n} B(t) \cosh \sqrt{\lambda_n} (t - \frac{b}{2}) - \sqrt{\lambda_n} A(t) \cosh \sqrt{\lambda_n} (t + \frac{b}{2}) = 1. \end{cases}$$

Solving this equation gives

$$\begin{cases} A(t) = \frac{1}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n} b} \sinh \sqrt{\lambda_n} (t - \frac{b}{2}) \\ B(t) = \frac{1}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n} b} \sinh \sqrt{\lambda_n} (t + \frac{b}{2}). \end{cases}$$

Therefore

$$g(y; t) = \frac{\sinh \sqrt{\lambda_n} (t - \frac{b}{2}) \sinh \sqrt{\lambda_n} (y + \frac{b}{2}) 1_{\{-\frac{b}{2} < y < t\}} + \sinh \sqrt{\lambda_n} (t + \frac{b}{2}) \sinh \sqrt{\lambda_n} (y - \frac{b}{2}) 1_{\{t < y < \frac{b}{2}\}}}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n} b}$$

and

$$\begin{aligned} \int_{-\infty}^{\infty} g(y; t) dy &= \int_{-\frac{b}{2}}^t \frac{\sinh \sqrt{\lambda_n} (y - \frac{b}{2})}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n} b} \sinh \sqrt{\lambda_n} (t + \frac{b}{2}) dy + \int_t^{\frac{b}{2}} \frac{\sinh \sqrt{\lambda_n} (y + \frac{b}{2})}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n} b} \sinh \sqrt{\lambda_n} (t - \frac{b}{2}) dy \\ &= \frac{\sinh \sqrt{\lambda_n} (y - \frac{b}{2})}{\lambda_n \sinh \sqrt{\lambda_n} b} \left[\cosh \sqrt{\lambda_n} (y + \frac{b}{2}) - 1 \right] + \frac{\sinh \sqrt{\lambda_n} (y + \frac{b}{2})}{\lambda_n \sinh \sqrt{\lambda_n} b} \left[1 - \cosh \sqrt{\lambda_n} (y - \frac{b}{2}) \right] \\ &= \frac{-\sinh \sqrt{\lambda_n} b - \sinh \sqrt{\lambda_n} (y - \frac{b}{2}) + \sinh \sqrt{\lambda_n} (y + \frac{b}{2})}{\lambda_n \sinh \sqrt{\lambda_n} b} \\ &= \frac{-\sinh \sqrt{\lambda_n} b + 2 \sinh \frac{\sqrt{\lambda_n} b}{2} \cosh \sqrt{\lambda_n} y}{\lambda_n \sinh \sqrt{\lambda_n} b} \\ &= -\frac{1}{\lambda_n} + \frac{\cosh \sqrt{\lambda_n} y}{\lambda_n \cosh \frac{\sqrt{\lambda_n} b}{2}}. \end{aligned}$$

Hence

$$\begin{aligned} Y_n(y) &= \frac{4}{n\pi} [(-1)^n - 1] \int_{-\infty}^{\infty} g(y; t) dt = \frac{4}{n\pi} [(-1)^{n+1} + 1] \left[\frac{a^2}{(n\pi)^2} - \frac{a^2 \cosh \frac{n\pi y}{a}}{(n\pi)^2 \cosh \frac{n\pi b}{2a}} \right] \\ &= \begin{cases} \frac{8a^2}{(n\pi)^3} \left[1 - \frac{\cosh \frac{n\pi y}{a}}{\cosh \frac{n\pi b}{2a}} \right] & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even.} \end{cases} \end{aligned}$$

Combined, we have

$$u(x, t) = \sum_{n=1}^{\infty} X_n(x) Y_n(y) = \frac{8a^2}{\pi^3} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3} \left[1 - \frac{\cosh \frac{2n+1}{a} \pi y}{\cosh \frac{2n+1}{2a} \pi b} \right] \sin \frac{2n+1}{a} \pi x.$$

(2)

证明. Similar to part (1), we find the solution to the eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(a) = 0 \end{cases}$$

as $X_n(x) = \sin \sqrt{\lambda_n} x$ with $\lambda_n = \left(\frac{n\pi}{a}\right)^2$ ($n \in \mathbb{N}$). Expanding $u(x, y)$ according to $\{X_n(x)\}_{n=1}^{\infty}$, we have

$$u(x, y) = \sum_{n=1}^{\infty} Y_n(y) X_n(x),$$

where

$$Y_n(y) = \frac{\int_0^a x^2 y X_n(x) dx}{\int_0^a X_n^2(x) dx} = \frac{2}{a} \int_0^a x^2 y X_n(x) dx = 2y \left\{ \frac{a^2 (-1)^{n+1}}{n\pi} + \frac{2a^2}{(n\pi)^3} [(-1)^n - 1] \right\}.$$

So the eigenvalue problem associated with $Y_n(x)$ becomes

$$\begin{cases} Y_n''(x) - \lambda_n Y_n(x) = 2y \left\{ \frac{a^2 (-1)^{n+1}}{n\pi} + \frac{2a^2}{(n\pi)^3} [(-1)^n - 1] \right\} \\ Y_n(-\frac{b}{2}) = Y_n(\frac{b}{2}) = 0. \end{cases}$$

The Green's function associated with $Y_n(y)$ is the same as that of part (1):

$$g(y; t) = \frac{\sinh \sqrt{\lambda_n} (t - \frac{b}{2}) \sinh \sqrt{\lambda_n} (y + \frac{b}{2}) 1_{\{-\frac{b}{2} < y < t\}} + \sinh \sqrt{\lambda_n} (t + \frac{b}{2}) \sinh \sqrt{\lambda_n} (y - \frac{b}{2}) 1_{\{t < y < \frac{b}{2}\}}}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n} b}.$$

Therefore

$$\begin{aligned} Y_n(y) &= \int_{-\infty}^{\infty} 2t \left\{ \frac{a^2 (-1)^{n+1}}{n\pi} + \frac{2a^2}{(n\pi)^3} [(-1)^n - 1] \right\} g(y; t) dt \\ &= \frac{2a^2}{n\pi} \left\{ (-1)^{n+1} + \frac{2}{(n\pi)^2} [(-1)^n - 1] \right\} \int_{-\infty}^{\infty} t g(y; t) dt. \end{aligned}$$

Note

$$\begin{aligned}
\int_{-\infty}^{\infty} tg(y; t)dt &= \frac{\sinh \sqrt{\lambda_n}(y + \frac{b}{2})}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n}b} \int_y^{\frac{b}{2}} t \sinh \sqrt{\lambda_n}(t - \frac{b}{2})dt + \frac{\sinh \sqrt{\lambda_n}(y - \frac{b}{2})}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n}b} \int_{-\frac{b}{2}}^y t \sinh \sqrt{\lambda_n}(t + \frac{b}{2})dt \\
&= \frac{\sinh \sqrt{\lambda_n}(y + \frac{b}{2})}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n}b} \left[\frac{b}{2\sqrt{\lambda_n}} - \frac{y}{\sqrt{\lambda_n}} \cosh \sqrt{\lambda_n}(y - \frac{b}{2}) + \frac{1}{\lambda_n} \sinh \sqrt{\lambda_n}(y - \frac{b}{2}) \right] + \\
&\quad \frac{\sinh \sqrt{\lambda_n}(y - \frac{b}{2})}{\sqrt{\lambda_n} \sinh \sqrt{\lambda_n}b} \left[\frac{b}{2\sqrt{\lambda_n}} + \frac{y}{\sqrt{\lambda_n}} \cosh \sqrt{\lambda_n}(y + \frac{b}{2}) - \frac{1}{\lambda_n} \sinh \sqrt{\lambda_n}(y + \frac{b}{2}) \right] \\
&= \frac{b[\sinh \sqrt{\lambda_n}(y + \frac{b}{2}) + \sinh \sqrt{\lambda_n}(y - \frac{b}{2})]}{2\lambda_n \sinh \sqrt{\lambda_n}b} + \frac{y \sinh(-b)\sqrt{\lambda_n}}{\lambda_n \sinh \sqrt{\lambda_n}b} \\
&= \frac{b}{2\lambda_n} \frac{\sinh \sqrt{\lambda_n}y}{\sinh \frac{\sqrt{\lambda_n}b}{2}} - \frac{y}{\lambda_n}.
\end{aligned}$$

Therefore

$$\begin{aligned}
Y_n(y) &= \frac{2a^2}{n\pi} \left\{ (-1)^{n+1} + \frac{2}{(n\pi)^2} [(-1)^n - 1] \right\} \left(\frac{b}{2\lambda_n} \frac{\sinh \sqrt{\lambda_n}y}{\sinh \frac{\sqrt{\lambda_n}b}{2}} - \frac{y}{\lambda_n} \right) \\
&= \begin{cases} -\frac{2a^2}{n\pi\lambda_n} \left(y - \frac{b}{2} \frac{\sinh \sqrt{\lambda_n}y}{\sinh \frac{\sqrt{\lambda_n}b}{2}} \right) + \frac{4}{(n\pi)^2} \frac{2a^2}{n\pi\lambda_n} \left(y - \frac{b}{2} \frac{\sinh \sqrt{\lambda_n}y}{\sinh \frac{\sqrt{\lambda_n}b}{2}} \right) & \text{if } n \text{ is odd,} \\ \frac{2a^2}{n\pi\lambda_n} \left(y - \frac{b}{2} \frac{\sinh \sqrt{\lambda_n}y}{\sinh \frac{\sqrt{\lambda_n}b}{2}} \right) & \text{if } n \text{ is even.} \end{cases}
\end{aligned}$$

Hence

$$\begin{aligned}
&u(x, y) \\
&= \sum_{n=1}^{\infty} Y_n(y) X_n(x) \\
&= \frac{2a^4}{\pi^3} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \left(y - \frac{b}{2} \frac{\sinh \frac{n\pi}{a}y}{\sinh \frac{n\pi}{2a}b} \right) \sin \frac{n\pi}{a}x + \frac{8a^4}{\pi^5} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^5} \left(y - \frac{b}{2} \frac{\sin \frac{2n+1}{a}\pi y}{\sinh \frac{2n+1}{2a}\pi b} \right) \sin \frac{2n+1}{a}\pi x.
\end{aligned}$$

注 26. The above solution differs from the textbook's answer by a sign. Check.

4. A slender beam is fixed at $x=0$, while the end at $x=l$ is subjected to a periodic force $Asin \omega t$. Assuming the initial displacement and initial velocity are both zero, solve the transverse vibration problem of the beam.

解: 设杆截面积 S , 杨氏模量 E

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0 \\ u|_{x=0} = 0 \quad \frac{\partial u}{\partial x}|_{x=l} = \frac{A \sin \omega t}{SE} \quad \text{令 } \frac{A}{SE} = b \\ u|_{t=0} = 0 \quad \frac{\partial u}{\partial t}|_{t=0} = 0 \end{cases}$$

$$\text{设 } u(x,t) = v(x) + w(x,t) \quad w(x,t) = \frac{A \sin \omega t}{SE} x = (b \sin \omega t) x$$

$$\text{设 } v(x,t) = f(x) \sin \omega t$$

$$\text{则 } f''(x) + \left(\frac{\omega^2}{a^2}\right) f(x) = 0 \Leftrightarrow v_{tt} - 0^2 v_{xx} = 0$$

$$\begin{cases} f(0) = 0 \\ f(l) = b \end{cases}$$

$$\Rightarrow f(x) = \frac{b a}{l} \frac{\sin(\omega x/a)}{\cos(\omega l/a)}$$

$$\begin{cases} \frac{\partial w}{\partial t} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 & 0 < x < l, t > 0 \\ w|_{x=0} = 0 \quad \frac{\partial w}{\partial x}|_{x=l} = 0 & t > 0 \\ w|_{t=0} = 0 \quad \frac{\partial w}{\partial t}|_{t=0} = -\frac{\partial v}{\partial t}|_{t=0} = -\frac{b a}{\cos(\omega l/a)} \sin \omega x & 0 \leq x \leq l \end{cases}$$

$$\text{设 } W(x,t) = X(x)T(t) \quad \frac{X''(x)}{X(x)} = \frac{T''(t)}{a^2 T(t)} = -\lambda$$

$$\Rightarrow X_n(x) = B_n \sin\left[\frac{(2n+1)\pi}{2l} x\right] \quad \lambda_n = \left(\frac{(2n+1)\pi}{2l}\right)^2 \quad n = 0, 1, 2, \dots$$

$$W(x,t) = \sum_{n=0}^{\infty} \left[C_n \sin \frac{(2n+1)\pi}{2l} x \cos \frac{(2n+1)\pi}{2l} a t + D_n \cos \frac{(2n+1)\pi}{2l} x \sin \frac{(2n+1)\pi}{2l} a t \right] \sin \frac{(2n+1)\pi}{2l} x \quad \text{代入得}$$

$$D_n = 0$$

$$\sum_{n=0}^{\infty} C_n \sin \frac{(2n+1)\pi}{2l} x \cdot \left(\frac{(2n+1)\pi}{2l} a t \right) \pi a = \frac{-b a}{\cos(\omega l/a)} \sin \frac{\omega x}{a}$$

$$\Rightarrow C_n = \frac{\frac{2}{l} \int_0^l \frac{-b a \sin \frac{\omega x}{a}}{\cos(\omega l/a)} \sin \frac{(2n+1)\pi}{2l} x dx}{\frac{(2n+1)\pi}{2l} \pi a} = (-1)^n \frac{16 b \omega l^2 a}{(2n+1)\pi} \cdot \frac{1}{(\omega l/a)^2 - [(2n+1)\pi a]^2}$$

$$\therefore u(x,t) = v(x,t) + w(x,t) = \frac{A a}{SE \omega} \cdot \frac{\sin(\omega x/a)}{\cos(\omega l/a)} \cdot \sin \omega t + \frac{4 A \omega}{SE \pi a} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)} \cdot \frac{\sin \frac{(2n+1)\pi}{2l} x x}{(\omega l/a)^2 - [(2n+1)\pi a]^2} \sin \frac{(2n+1)\pi}{2l} a t \cdot \sin \frac{(2n+1)\pi}{2l} x$$

若 ω 恰为杆个固有频率 ($\omega = \omega_n = (2n+1)\pi a / 2l$)

则将 $n=m$ 项与初化函数合并, 再用洛必达求极限,

$$u(x,t) = \frac{6 A l}{SE} \cdot \frac{(-1)^m}{(2m+1)\pi} \cdot \sin \frac{(2m+1)\pi}{2l} x x \sin \frac{(2m+1)\pi}{2l} a t + \frac{2 A}{SE} \cdot \frac{(-1)^m}{(2m+1)\pi} \left[x \cos \frac{(2m+1)\pi}{2l} x \sin \frac{(2m+1)\pi}{2l} a t + a t \sin \frac{(2m+1)\pi}{2l} x \cos \frac{(2m+1)\pi}{2l} a t \right]$$

$$+ \frac{8(2m+1) A l}{S \pi^2 E} \cdot \frac{(-1)^m}{\pi a} \cdot \frac{1}{(2m+1)^2 - (2m+1)^2} \sin \frac{(2m+1)\pi}{2l} x x \sin \frac{(2m+1)\pi}{2l} a t$$

$\sum_{n=0}^{\infty}$ 表示和式不含 $n=m$ 项

5. Try to solve the following solution problem

$$\frac{\partial^2 u}{\partial t^2} - a^2 \frac{\partial^2 u}{\partial x^2} = 0,$$

$$u|_{x=0} = \cos \frac{\pi}{l} a t, \quad \frac{\partial u}{\partial x} \Big|_{x=l} = 0,$$

$$u|_{t=0} = \cos \frac{\pi}{l} x, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = \sin \frac{\pi}{2l} x.$$

Proof. We choose $v(x,t) = \cos \frac{\pi}{l} x \cos \frac{\pi}{l} a t$ and suppose $u(x,t) = v(x,t) + w(x,t)$. Then $w(x,t)$ satisfies the equation

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} - a^2 \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, & \frac{\partial w}{\partial x} \Big|_{x=l} = 0 \\ w(x,0) = 0, & \frac{\partial w}{\partial t} \Big|_{t=0} = \sin \frac{\pi}{2l} x. \end{cases}$$

Solving the associated eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = 0, X'(l) = 0 \end{cases}$$

gives the eigenfunctions $\{X_n(x)\} = \{\sin \sqrt{\lambda_n} x\}$, with $\lambda_n = \left(\frac{\pi+2n\pi}{2l}\right)^2$. Suppose $w(x,t) = \sum_{n=0}^{\infty} T_n(t) X_n(x)$, then we must have

$$\begin{cases} \sum_{n=0}^{\infty} (T_n''(t) + \lambda_n a^2) X_n(x) = 0 \\ \sum_{n=0}^{\infty} T_n(0) X_n(x) = 0, \sum_{n=0}^{\infty} T_n'(0) X_n(x) = X_0(x). \end{cases}$$

Therefore $T_n(t) \equiv 0$ for $n \geq 1$ and $T_0(t) = \frac{2l}{\pi a} \sin \frac{\pi}{2l} a t$. Combined, we conclude $u(x,t) = \cos \frac{\pi}{l} x \cos \frac{\pi}{l} a t + \frac{2l}{\pi a} \sin \frac{\pi}{2l} x \sin \frac{\pi}{2l} a t$. $\square$

6. Try to solve the following solution problem

$$\frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0,$$

$$u|_{x=0} = A e^{j\omega t}, \quad u|_{x=l} = 0,$$

$$u|_{t=0} = 0.$$

解: 令 $u(x, t) = V(x, t) + W(x, t)$. $V(x, t)$ 为特解. 设为 $V(x, t) = A(x) e^{j\omega t}$.

$$\text{代入方程得 } j\omega A(x) e^{j\omega t} - \kappa e^{j\omega t} A''(x) = 0.$$

$$A''(x) - \frac{j\omega}{\kappa} A(x) = 0. \quad A(0) = A, \quad A(l) = 0.$$

$$\text{设 } A(x) = k_1 \sinh\left(\sqrt{\frac{j\omega}{\kappa}} x\right) + k_2 \cosh\left(\sqrt{\frac{j\omega}{\kappa}} x\right), \quad A(0) = k_2 = A.$$

$$A(l) = k_1 \sinh\left(\sqrt{\frac{j\omega}{\kappa}} l\right) + A \cosh\left(\sqrt{\frac{j\omega}{\kappa}} l\right) = 0 \quad k_1 = \frac{-A \cosh\left(\sqrt{\frac{j\omega}{\kappa}} l\right)}{\sinh\left(\sqrt{\frac{j\omega}{\kappa}} l\right)}$$

$$\therefore V(x, t) = \left[\frac{-A \cosh\left(\sqrt{\frac{j\omega}{\kappa}} l\right)}{\sinh\left(\sqrt{\frac{j\omega}{\kappa}} l\right)} \sinh\left(\sqrt{\frac{j\omega}{\kappa}} x\right) + A \cosh\left(\sqrt{\frac{j\omega}{\kappa}} x\right) \right] e^{j\omega t}.$$

$$\Delta W(x, t) = X(x) \cdot T(t). \quad W|_{x=0} = 0, \quad W|_{x=l} = 0, \quad W|_{t=0} = -V(x, 0)$$

$$X(x) T'(t) - \kappa X''(x) T(t) = 0 \quad \frac{X''(x)}{X(x)} = \frac{1}{\kappa} \frac{T'(t)}{T(t)} = -\lambda. \quad T'(x) + \lambda \kappa T(x) = 0.$$

$$\text{本征值问题: } X''(x) + \lambda X(x) = 0. \quad X(0) = 0, \quad X(l) = 0.$$

$$\lambda = \left(\frac{n\pi}{l}\right)^2, \quad X(x) = \sin\left(\frac{n\pi}{l} x\right), \quad n = 1, 2, \dots$$

$$\cdot T_n(t) = C_n e^{-\lambda \kappa t} = C_n e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t} \quad W_n(x, t) = \sin\left(\frac{n\pi}{l} x\right) \cdot C_n e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t}$$

$$W|_{t=0} = C_n \sin\left(\frac{n\pi}{l} x\right) = -k_1 \sinh\left(\sqrt{\frac{j\omega}{\kappa}} x\right) - k_2 \cosh\left(\sqrt{\frac{j\omega}{\kappa}} x\right)$$

$$\sum_{n=1}^{\infty} C_n \int_0^l \sin\left(\frac{n\pi}{l} x\right) \sin\left(\frac{m\pi}{l} x\right) dx = -k_1 \int_0^l \sinh\left(\sqrt{\frac{j\omega}{\kappa}} x\right) \sin\left(\frac{m\pi}{l} x\right) dx - k_2 \int_0^l \cosh\left(\sqrt{\frac{j\omega}{\kappa}} x\right) \sin\left(\frac{m\pi}{l} x\right) dx.$$

$$C_m = k_1 \frac{2m\pi\kappa \cdot (-1)^m \sinh\left(\sqrt{\frac{j\omega}{\kappa}} l\right)}{(m\pi)^2 \kappa + j\omega l^2} - k_2 \frac{2m\pi\kappa}{(m\pi)^2 \kappa + j\omega l^2} [1 - (-1)^m \cosh\left(\sqrt{\frac{j\omega}{\kappa}} l\right)]$$

$$\text{代入 } k_1, k_2 \quad C_m = -\frac{2m\pi\kappa A}{(m\pi)^2 \kappa + j\omega l^2}$$

$$\therefore W(x, t) = \sum_{n=1}^{\infty} \frac{-2n\pi\kappa A}{(n\pi)^2 \kappa + j\omega l^2} \sin\left(\frac{n\pi}{l} x\right) \cdot e^{-\left(\frac{n\pi}{l}\right)^2 \kappa t}.$$

$$\Delta u(x, t) = V(x, t) + W(x, t) = \dots$$

$$W|_{t=0} = -A \frac{\sinh (1+i) \alpha (l-x)}{\sinh (1+i) \alpha l}$$

$$u = X(x)T(t), \quad XT' - k \cdot X''T = 0$$

$$\cos(iz) = \cosh z$$

$$-i \sin(iz) = \sinh z$$

$$\frac{X''}{X} = \frac{T'}{kT} = -\lambda$$

$$\begin{cases} X'' + \lambda X = 0 \\ T' + \lambda k T = 0 \end{cases} \quad \begin{cases} X(0) = 0 \\ X(l) = 0 \end{cases}$$

$$\frac{e^{-x} - e^x}{2i} = i \sinh x$$

$\lambda = 0$ 不是特征值.

$$B = 0, A \neq 0, \sinh \sqrt{\lambda} x = 0, \lambda = \left(\frac{n\pi}{l}\right)^2, n = 1, 2, 3, \dots$$

$$X_n(x) = \sin \frac{n\pi}{l} x$$

$$T_n(t) = C_n e^{-\left(\frac{n\pi}{l}\right)^2 k t}$$

$$W|_{t=0} = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{l} x = -A \frac{\sinh (1+i) \alpha (l-x)}{\sinh (1+i) \alpha l}$$

$$= -A \frac{\sinh (i-1) \alpha (l-x)}{\sinh (i-1) \alpha l}$$

$$C_n = \frac{2}{l} \int_0^l -A \frac{\sinh [(i-1) \alpha (l-x)] \sin \frac{n\pi}{l} x}{\sinh (i-1) \alpha l} dx$$

$$C_n = \frac{2}{l} \int_0^l -A \frac{\sin[(i-1)\alpha(l-x)] \sin \frac{n\pi}{l} x}{\sin(i-1)\alpha} dx. \quad 2A$$

$$\begin{aligned} & \int_0^l 2 \sin(\omega(l-x)) \sin \frac{n\pi}{l} x \, dx \quad 4 \\ &= \int_0^l \cos[\omega l - \omega x - \frac{n\pi}{l} x] - \cos[\omega l - \omega x + \frac{n\pi}{l} x] \, dx \\ &= \frac{1}{\omega' + \frac{n\pi}{l}} \sin[\omega l - \omega x - \frac{n\pi}{l} x]_0^l - \frac{1}{\omega' - \frac{n\pi}{l}} \sin[\omega l - \omega x + \frac{n\pi}{l} x]_0^l \\ &= \frac{1}{\omega' - \frac{n\pi}{l}} \sin(\omega l) - \frac{1}{\omega' + \frac{n\pi}{l}} \sin(\omega l) \\ &= \frac{2n\pi/l}{\omega'^2 - \frac{n^2\pi^2}{l^2}} \sin(\omega l) \end{aligned}$$

$$\begin{aligned} C_n &= -\frac{1}{l} \frac{1}{\sin(i-1)\alpha} \frac{2n\pi/l}{(i-1)^2\alpha^2 - \frac{n^2\pi^2}{l^2}} \sin(i-1)\alpha. \\ &= -\frac{2A n\pi}{-2i \frac{\omega}{2k} l^2 - n^2\pi^2} = \frac{2kA n\pi}{(n^2\pi^2 k + i\omega l^2)} \end{aligned}$$

$$u(x,t) = A \frac{\sinh[(i+1)\sqrt{\frac{\omega}{2k}}(l-x)]}{\sinh[(i+1)\sqrt{\frac{\omega}{2k}}l]} e^{i\omega t} - 2kA \sum_{n=1}^{\infty} \frac{n\pi}{(n^2\pi^2 k + i\omega l^2)} \sin \frac{n\pi}{l} x e^{-(\frac{n\pi}{l})^2 kt}$$

7. Try to solve the following solution problem

$$\frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0,$$

$$u|_{x=0} = A \exp\{-\alpha^2 \kappa t\}, \quad u|_{x=l} = B \exp\{-\beta^2 \kappa t\},$$

$$u|_{t=0} = 0.$$

Proof. We choose $v(x, t) = \frac{l-x}{l} A e^{-\alpha^2 \kappa t} + \frac{x}{l} B e^{-\beta^2 \kappa t}$ and suppose $u(x, t) = v(x, t) + w(x, t)$. Note $v(0, t) = A e^{-\alpha^2 \kappa t}$ and $v(l, t) = B e^{-\beta^2 \kappa t}$, we have the following equation for $w(x, t)$:

$$\begin{cases} \frac{\partial w}{\partial t} - \kappa \frac{\partial^2 w}{\partial x^2} = - \left[\frac{\partial v}{\partial t} - \kappa \frac{\partial^2 v}{\partial x^2} \right] = \frac{l-x}{l} A \alpha^2 \kappa e^{-\alpha^2 \kappa t} + \frac{x}{l} B \beta^2 \kappa e^{-\beta^2 \kappa t} = f(x, t) \\ w|_{x=0} = w|_{x=l} = 0 \\ w(x, 0) = -v(x, 0) = -\frac{l-x}{l} A - \frac{x}{l} B = \phi(x). \end{cases}$$

Solving the eigenvalue problem

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

gives the eigenfunctions $\{X_n(x)\}_{n=1}^{\infty}$, where $X_n(x) = \sin \sqrt{\lambda_n} x$ with $\lambda_n = \left(\frac{n\pi}{l}\right)^2$. Expand $f(x, t)$ according to $\{X_n(x)\}$, we have $f(x, t) = \sum_{n=1}^{\infty} g_n(t) X_n(x)$ with

$$g_n(t) = \frac{2}{l} \int_0^l \left[\frac{l-x}{l} A \alpha^2 \kappa e^{-\alpha^2 \kappa t} + \frac{x}{l} B \beta^2 \kappa e^{-\beta^2 \kappa t} \right] X_n(x) dx.$$

Note $\int_0^l X_n(x) dx = \frac{1}{-\sqrt{\lambda_n}} \cos \sqrt{\lambda_n} x \Big|_{x=0}^l = \frac{(-1)^n - 1}{-\frac{n\pi}{l}} = \frac{l((-1)^{n+1} + 1)}{n\pi}$ and

$$\int_0^l x X_n(x) dx = \frac{1}{-\sqrt{\lambda_n}} \int_0^l x d \cos \sqrt{\lambda_n} x = \frac{l(-1)^n}{-\frac{n\pi}{l}} = \frac{l^2}{n\pi} (-1)^{n+1}.$$

Therefore

$$\begin{aligned} g_n(t) &= \frac{2}{l} \int_0^l \left[A \alpha^2 \kappa e^{-\alpha^2 \kappa t} X_n(x) + \frac{B \beta^2 \kappa e^{-\beta^2 \kappa t} - A \alpha^2 \kappa e^{-\alpha^2 \kappa t}}{l} x X_n(x) \right] dx \\ &= \frac{2\kappa}{n\pi} A \alpha^2 e^{-\alpha^2 \kappa t} + \frac{2\kappa}{n\pi} B \beta^2 e^{-\beta^2 \kappa t} (-1)^{n+1}. \end{aligned}$$

Expand $\phi(x)$ according to $\{X_n(x)\}_{n=1}^{\infty}$, we have $\phi(x) = \sum_{n=1}^{\infty} a_n X_n(x)$ where

$$a_n = \frac{2}{l} \int_0^l \left[-A + \frac{A-B}{l} x \right] X_n(x) dx = -\frac{2A}{n\pi} - \frac{2B}{n\pi} (-1)^{n+1}.$$

If we let $w(x, t) = \sum_{n=1}^{\infty} T_n(t) X_n(x)$, then

$$\begin{cases} \sum_{n=1}^{\infty} [T'_n(t) + \kappa \lambda_n T_n(t)] X_n(x) = \sum_{n=1}^{\infty} g_n(t) X_n(x) \\ \sum_{n=1}^{\infty} T_n(0) X_n(x) = \sum_{n=1}^{\infty} a_n X_n(x). \end{cases}$$

By orthogonality of $\{X_n(x)\}$, we have

$$\begin{cases} T'_n(t) + \kappa \lambda_n T_n(t) = g_n(t) \\ T_n(0) = a_n, \end{cases}$$

which implies

$$\begin{aligned} T_n(t) &= e^{-\kappa \lambda_n t} \int_0^t g_n(s) e^{\kappa \lambda_n s} ds + a_n \\ &= \frac{2\kappa}{n\pi} \left[A \alpha^2 \frac{e^{-\alpha^2 \kappa t} - e^{-\kappa \lambda_n t}}{-\alpha^2 \kappa + \kappa \lambda_n} + B \beta^2 (-1)^{n+1} \frac{e^{-\beta^2 \kappa t} - e^{-\kappa \lambda_n t}}{-\beta^2 \kappa + \kappa \lambda_n} \right] + a_n \end{aligned}$$

Combined, we can write the solution $u(x, t)$ to the original problem as

$$\left[\frac{l-x}{l} A e^{-\alpha^2 \kappa t} + \frac{x}{l} B e^{-\beta^2 \kappa t} \right] + \sum_{n=1}^{\infty} \left\{ \frac{2\kappa}{n\pi} \left[A \alpha^2 \frac{e^{-\alpha^2 \kappa t} - e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t}}{-\alpha^2 \kappa + \kappa \left(\frac{n\pi}{l}\right)^2} + B \beta^2 (-1)^{n+1} \frac{e^{-\beta^2 \kappa t} - e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t}}{-\beta^2 \kappa + \kappa \left(\frac{n\pi}{l}\right)^2} \right] + a_n \right\} \sin \frac{n\pi}{l} x.$$

Remark 27. If we choose $v(x, t) = A \frac{\sin \alpha(l-x)}{\sin \alpha l} e^{-\alpha^2 \kappa t} + B \frac{\sin \beta x}{\sin \beta l} e^{-\beta^2 \kappa t}$, then it's easy to verify $\frac{\partial v}{\partial t} - \kappa \frac{\partial^2 v}{\partial x^2} = 0$. So this choice of $v(x, t)$ simultaneously homogenizes boundary condition and the differential equation, which makes the solution much easier.

$$v(x, t) = A \frac{\sin \alpha(l-x)}{\sin \alpha l} e^{-\alpha^2 \kappa t} + B \frac{\sin \beta x}{\sin \beta l} e^{-\beta^2 \kappa t}。$$

$$\text{令 } u = v + w, \text{ 则 } \begin{cases} \frac{\partial w}{\partial t} - \kappa \frac{\partial^2 w}{\partial x^2} = 0 \\ w|_{x=0} = 0, w|_{x=l} = 0 \\ w|_{t=0} = -A \frac{\sin \alpha(l-x)}{\sin \alpha l} - B \frac{\sin \beta x}{\sin \beta l} \end{cases}。$$

$$w = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{l} x e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t},$$

$$\begin{aligned} C_n &= -\frac{2A}{l \sin \alpha l} \int_0^l \sin \alpha(l-x) \sin \frac{n\pi}{l} x dx - \frac{2B}{l \sin \beta l} \int_0^l \sin \beta x \sin \frac{n\pi}{l} x dx \\ &= -\frac{2A}{l} \int_0^l \cos \alpha x \sin \frac{n\pi}{l} x dx + \frac{2A}{l} \cot \alpha l \int_0^l \sin \alpha x \sin \frac{n\pi}{l} x dx - \frac{2B}{l \sin \beta l} \int_0^l \sin \beta x \sin \frac{n\pi}{l} x dx \\ &= -\frac{2An\pi}{(n\pi)^2 - (\alpha l)^2} \left[1 - (-1)^n \cos \alpha l \right] - \frac{2An\pi(-1)^n}{(n\pi)^2 - (\alpha l)^2} \cos \alpha l + \frac{2(-1)^n n\pi B}{(n\pi)^2 - (\beta l)^2} \\ &= -\frac{2An\pi}{(n\pi)^2 - (\alpha l)^2} + \frac{2(-1)^n n\pi B}{(n\pi)^2 - (\beta l)^2}。 \end{aligned}$$

上面假设不存在正整数 n, m , 使 $n\pi = \alpha l$, $m\pi = \beta l$ 。

$$\begin{aligned} u(x, t) &= A \frac{\sin \alpha(l-x)}{\sin \alpha l} e^{-\alpha^2 \kappa t} + B \frac{\sin \beta x}{\sin \beta l} e^{-\beta^2 \kappa t} \\ &\quad - \sum_{n=1}^{\infty} 2n\pi \left[\frac{A}{(n\pi)^2 - (\alpha l)^2} - \frac{(-1)^n B}{(n\pi)^2 - (\beta l)^2} \right] \sin \frac{n\pi}{l} x e^{-\kappa \left(\frac{n\pi}{l}\right)^2 t}。 \end{aligned}$$